



# OMERTA AI

## Owner's Manual

*Silence Is The Only Unbreakable Code*

A coding, systems and firmware agent that runs on your own hardware and asks before it acts. Android without Termux, Linux, macOS, Windows.

You supply the model; OMERTA supplies everything around it.

**Version 1.16.0**

read chapter 2 first - it explains what you must supply

## CHAPTER 1

# What OMERTA AI Is

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OMERTA AI is an agent that runs on hardware you own. It reads and writes your code, drives a real shell, inspects and builds firmware images, remembers what you teach it, and asks your permission before it changes anything.

## The one-sentence version

It is the whole machine around a language model — the memory, the tools, the terminal, the sandbox, the approval gate, the interface — running entirely on your device, with the model itself being a part you choose and plug in.

## What makes it different

- **It asks first.** Every action the model wants to take is classified and shown to you before it happens. Destructive and irreversible operations always ask, no matter how you set the policy.
- **It runs on your hardware.** The Android build carries its own Python interpreter. There is no Termux, no companion app, no bootstrap download, and no server of ours in the path.
- **It has a real terminal.** A genuine PTY-backed shell with job control, plus 288 bundled commands, because Android's own userland is too thin to work in.
- **It does not guess.** The firmware side in particular labels every value CONFIRMED, LIKELY, INFERRED or UNKNOWN, and refuses to turn an UNKNOWN into a guess to keep a task moving.
- **It is yours to shape.** Projects, branched and queued chats, passcode-locked private chats, a learning shelf you can add to and take away from, and a theme system down to the fonts and the icon.

## Who it is for

Someone who wants an agent on a phone or a laptop that behaves like a capable colleague rather than a toy: one that can be handed a real repository, a real device tree or a real shell, and that will tell you what it does not know instead of inventing it.

## CHAPTER 2

## The Model — Read This First

The single most important thing to understand about OMERTA: it does not contain a language model, and it will not think without one.

**OMERTA AI is not itself an AI model.** It is an agent — the harness. Claude, GPT and the rest are models: the part that reasons. OMERTA is everything else. Installed and opened with nothing configured, it will start, show its interface, run its terminal and browse your files, and it will not answer a single question, because there is nothing behind the chat box yet.

### So what do I have to supply?

Exactly one of these two things:

#### 1. An API key, for a model in the cloud

Paste it into Settings. The key is stored on your device and sent only to that provider. This is the easiest route and gives you the strongest models, at the cost of needing a connection and of your prompts leaving the device.

| Provider           | Setting            | Notes                                     |
|--------------------|--------------------|-------------------------------------------|
| Claude (Anthropic) | ANTHROPIC_API_KEY  | Sonnet and Opus; the default choice       |
| OpenAI             | OPENAI_API_KEY     | GPT and o-series                          |
| OpenRouter         | OPENROUTER_API_KEY | Many models behind one key                |
| Groq               | GROQ_API_KEY       | Very fast, open-weight models             |
| Google Gemini      | GEMINI_API_KEY     | Gemini via its OpenAI-compatible endpoint |

#### 2. A local model server, for no key and no internet

OMERTA speaks to Ollama, llama.cpp and LM Studio. Point it at the host and port in Settings. Nothing leaves your network, there is no key, no per-message cost and no usage limit.

| Server    | Default address          | Getting started                              |
|-----------|--------------------------|----------------------------------------------|
| Ollama    | http://127.0.0.1:11434   | ollama serve && ollama pull qwen2.5-coder:7b |
| llama.cpp | http://127.0.0.1:8080    | llama-server -m your-model.gguf              |
| LM Studio | http://127.0.0.1:1234/v1 | Enable the local server in its UI            |

### The honest caveat on phones

Those local addresses say 127.0.0.1 — this device. On a laptop that is easy: run Ollama on the same machine. **On a phone there is currently nothing to run**, because OMERTA does not yet bundle an inference engine. So on Android, "offline mode" today means *pointing at a model server on your own network* — your PC at 192.168.x.x running Ollama — not a model running on the handset.

That is the one gap between what OMERTA is and the phrase "runs on its own without an API key". Closing it means shipping a llama.cpp server inside the APK and letting you load a GGUF file. It is the first recommendation in Chapter 12.

## Online, offline and auto

| Mode    | What it does                                                                             |
|---------|------------------------------------------------------------------------------------------|
| online  | Cloud providers only. Never falls back to a local model.                                 |
| offline | Local providers only. Never touches the network — verified by test, not merely intended. |
| auto    | Prefers your chosen provider and falls back through the routing order.                   |

## CHAPTER 3

## Installing

One agent, five shapes. Pick whichever matches the machine in front of you; they share the same code, the same approval gate and the same data format.

### Android

Copy the APK to the phone and open it; allow "install unknown apps" for whatever you opened it from. Everything is inside: CPython, the agent, the web interface and the command-line toolset. It needs no network to start.

Updates install straight over the previous version and keep your chats, projects, themes and learned files. **Do not uninstall first** — that deletes all of it.

### Desktop

| Format    | Install                                                                                              |
|-----------|------------------------------------------------------------------------------------------------------|
| ApplImage | <code>chmod +x OMERTA-AGENT-desktop-*.ApplImage &amp;&amp; ./OMERTA-AGENT-desktop-*.ApplImage</code> |
| .deb      | <code>sudo apt install ./omerta-agent-desktop*_amd64.deb</code>                                      |
| tar.gz    | Unpack anywhere and run the binary inside. No install step.                                          |

### Command line

| Format            | Install                                               | Needs                          |
|-------------------|-------------------------------------------------------|--------------------------------|
| CLI ApplImage     | <code>chmod +x</code> and run                         | nothing — carries CPython 3.12 |
| Standalone binary | <code>chmod +x omerta-linux-x86_64</code>             | nothing                        |
| .deb (small)      | <code>sudo apt install ./omerta-agent*_all.deb</code> | system python3                 |
| pip               | <code>pip install omerta_agent-*.whl</code>           | Python 3.9+ (incl. Termux)     |

Use `apt install ./file.deb` rather than `dpkg -i` for the small .deb: dpkg alone will not pull python3-requests and python3-yaml, and leaves the package unconfigured.

### Confirming which build you are running

Every build carries a stamp naming the commit that produced it. In the app it is the box at the top of Settings — tap to copy. On a terminal, `omerta --version`. Over HTTP, `GET /api/version`. If a copy is unstamped it says so rather than claiming a version it cannot prove.

## CHAPTER 4

## The Approval Gate

The rule that everything else is built around: the model proposes, you dispose.

Every action the agent wants to take — run a command, write a file, flash an image — is classified before it happens and shown to you with the exact command it intends to run. Nothing is paraphrased.

### The four tiers

| Tier      | Examples                                                                            | Behaviour                                                                                                  |
|-----------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| DENY      | <code>rm -rf /</code> , wiping a disk, fork bombs                                   | Refused outright. Cannot be approved, because the mistake is unrecoverable and it is nearly always a typo. |
| HIGH RISK | flashing firmware, force push, git reset --hard, deleting a branch, package removal | Always asks, on every policy setting, including the most permissive.                                       |
| NORMAL    | building, running tests, editing files, committing                                  | Asks, unless you have set the policy to trust this class.                                                  |
| LOW RISK  | reading files, listing directories, status queries                                  | Runs without interrupting you.                                                                             |

### The policy dial

In Settings, or the chip at the top of the screen. It moves where the line sits for NORMAL actions only — **DENY** and **HIGH RISK** do not move, whatever you choose.

| Setting     | Meaning                                                    |
|-------------|------------------------------------------------------------|
| asks first  | The default. Anything that changes state asks.             |
| build trust | Routine build and test commands run; edits still ask.      |
| trusted     | Normal work proceeds; high-risk and destructive still ask. |

### What the gate does not cover, and why

The terminal, the editor's saves, the sandbox and loading a file by path do not prompt. Those are *you* operating your own device, not the model acting. Prompting for each keystroke would be absurd. That is also exactly why those four surfaces refuse any request that did not come from this device — see Chapter 10.

## CHAPTER 5

## The Interface

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Eight tabs. On the phone this is Jetpack Compose — real Android widgets, no WebView and no HTML. The desktop app and a browser on your network reach the same routes.

### CHAT

Where the work happens. No message limit and no truncation of your conversation. The agent will argue with you: if it thinks your approach is wrong it will say so and offer an alternative — and then do what you decided.

Work modes change how it approaches a request:

| Mode       | Behaviour                                                  |
|------------|------------------------------------------------------------|
| plan       | Produces a plan and stops. Writes nothing.                 |
| brainstorm | Options and trade-offs, deliberately divergent.            |
| research   | Reads and reports. Investigates without changing anything. |
| build      | Executes, still through the approval gate.                 |
| debate     | Argues both sides before settling on one.                  |

- **Type-ahead** — a message typed while a turn is running is held and sent in order, not dropped. **STOP** cancels the turn in flight and keeps whatever was already said.
- **Tap any message** for copy, pin, redo or edit. **Pin** puts it in that project's system prompt from the next turn on; **edit** drops everything after it and runs again from there.
- **Cost** — each reply carries its own token and spend estimate. The tilde is not decoration: most providers do not report usage, so the figure is derived from text length.
- **Files named in a reply** — a stack trace or a diff header becomes a button that opens that file in CODE.
- **Dictation** asks the platform recogniser to prefer a language pack already on the device. It is a preference, not a guarantee, and nothing in the app can verify which way it went.

### TERMINAL

A real shell with a real controlling terminal — job control, Ctrl-C, and full-screen programs such as vi all work. The banner tells you the shell, whether it is a PTY, whether the deny-guard is on, and how many tools are available. See Chapter 6.

### CODE

A file browser and editor over your project, with search across the indexed tree by text or by symbol, line numbers and syntax highlighting.

A save is never direct. **Review & save** describes the write and returns a diff having written nothing; only confirming that diff moves the file, and a backup of the previous contents is kept as it goes. Earlier copies of the open file are

listed underneath it, and restoring one loads it into the editor rather than writing it — so a restore passes the same diff you would see for any other change. That is the contract the agent itself is held to, which is what makes a change made by hand and a change made by the model equally recoverable.

## GIT

Branch, the counts of staged, unstaged and untracked files, the working and staged diff, recent commits and branches.

Everything on this screen is **read-only, and that is the feature**. The module behind it enforces a whitelist of git subcommands that cannot change a repository. The buttons that would change one — stage, commit, push — hand the request to the agent instead, so the exact command appears on an approval card before it runs. Two ways to move a branch, one of them ungated, is how a safety property quietly stops being one.

## TOOLS

- **Another device** — adb over TCP: connect, read properties, run a shell command, and show this device's public key for the other one to accept.
- **Sandbox** — a copy of the project that can be built and run for real. Nothing reaches the real tree until you accept it.
- **Memory** — search what it remembers, and read the preferences it inferred from what you approved and refused.
- **Cost** — calls, tokens and estimated spend over 7, 30 or 90 days, with the backend's own caveats about which figures are estimated and which could not be priced.
- **Sync and pairing** — pull memory and chats from another OMERTA on your network. Pairing shows a short code for two minutes; see Chapter 10.
- **Backup** — an encrypted archive of everything, with a dry restore that reports what it would overwrite before it does.
- **Audit log** — every command that touched the device, whether the agent proposed it or you typed it.

## CHATS

- **Projects** — separate workspaces. Each has its own memory, its own learned documents and its own conversation history.
- **Branching** — fork a conversation at any message to try a different direction without losing the original.
- **Queue** — line up several messages; each runs as the previous one finishes, so you can leave it working.
- **Private chats** — locked behind a passcode and encrypted on disk with a key derived from it. Forget the passcode and the contents are gone; there is no recovery, which is the point.
- **Archive, export and import** — conversations are yours to move between devices.

## LEARN

Upload documents, code or notes for the agent to study. It indexes them and uses them as background for that project. Crucially you can **remove** what it learned — individual documents and individual learned behaviours —

and the removal is real: the row is deleted and verified gone, not merely hidden.

## SETTINGS

Provider and model, network mode, approval policy, API keys, the on-device engine, weights, themes, the terminal guard, sandbox containment, appearance, and the build stamp. Every writable setting is rendered from the backend's own description of itself, so a setting added later appears without the app being changed — and the search box at the top is there because there are now more than twenty.

- **API keys, per project** — a key can be stored for everything, or for one project alone. A project with its own keys does **not** fall back to the shared one if its key is removed: it was moved off that account deliberately, and quietly putting it back would be the failure this prevents.
- **Scheduled turns** — "every morning, check the build". A scheduled turn runs under the same approval policy as one you typed: it stops at the gate and the approval waits for you. The clock is a thread inside the app, not cron — a phone that kills OMERTA kills the schedule with it, and a missed run happens at the next start and says it was late.
- **Audit chain** — seals the log so that each entry's digest includes the one before it. See Chapter 10 for what that does and does not prove.
- **App lock** — the device's own passcode in front of OMERTA. Deliberately the system credential, not a second secret of OMERTA's own kept on the same device.
- **Last crash** — the stack trace, the device, its ABIs and the build, ready to copy or share. A crash that only exists in logcat needs a cable and a laptop to read.
- **Panic wipe** — everything this app holds, on a phrase typed in full. See Chapter 10.

## CHAPTER 6

## The Terminal and Its Toolset

Android gives an app a very thin userland. OMERTA brings its own.

`/system/bin` on Android is `toybox`, and it is missing most of what a terminal is actually for: no `wget`, no `awk`, no `vi`, no `less`, no `unzip`, no `tar` that can write. So the APK ships a static BusyBox and puts **288 commands** on your `PATH`.

### What you get

`wget tar unzip gzip bzip2 xz cpio sed awk grep find xargs diff patch vi less more sort uniq cut tr wc head tail nc ping traceroute nslookup ip ifconfig netstat route ps top kill df du stat strings xxd sha256sum md5sum base64 bc dc tree watch crond blkid fdisk losetup mount` — and roughly 240 more.

A full list is written to `T00LS.txt` beside the tool directory on the device; read it with `less ../T00LS.txt` from the `bin` directory.

### Deliberate omissions

| Not linked                                                                                              | Why                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>su</code> , <code>login</code> , <code>passwd</code>                                              | They need <code>setuid</code> root. On a rooted phone, linking these would shadow the real <code>/system/bin/su</code> and break root — use that one. On an unrooted phone there is no working <code>sudo</code> at all, and shipping one that always fails would be worse than not shipping it. |
| <code>sh</code> , <code>ash</code>                                                                      | The session shell is chosen deliberately elsewhere. Run <code>ash</code> by name if you want BusyBox's.                                                                                                                                                                                          |
| <code>init</code> , <code>reboot</code> , <code>halt</code> , <code>chroot</code> , <code>insmod</code> | Require privileges no Android app has.                                                                                                                                                                                                                                                           |

### Two limits worth knowing

**HTTPS is not authenticated.** BusyBox's TLS does not validate certificates and will tell you so. `wget https://...` works, but treat it as unauthenticated transport: fine for fetching something you are going to checksum, not fine for anything secret. Ask the agent to fetch instead when it matters — it uses Python's TLS, which does validate.

**There is no python3 on the command line.** The app embeds CPython in its own process rather than as a separate executable, so there is no interpreter for the shell to run. Use the `CODE` tab, or ask the agent to run Python for you.

### The deny-guard

The hard-deny list applies in the terminal too: `rm -rf /` is refused even though you typed it yourself. Matching is argument-precise, so `rm -rf /tmp/build` is not caught by the rule against `rm -rf /`. Set `OMERTA_TERM_GUARD=0` if you genuinely need one. Every command is logged either way.

## CHAPTER 7

## The Sandbox

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Somewhere to run something you do not trust yet.

A scratch sandbox builds and executes code exactly as it would for real, but nothing it produces reaches your project until you say so. You can keep the result, throw it away, or keep part of it.

### Two different protections

| Layer              | Protects                           | How                                                                                                                                                     |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scratch workspace  | your project from your experiment  | Writes are confined to the sandbox directory and merged only on your confirmation.                                                                      |
| Device containment | your device from a hostile program | bubblewrap namespaces: no network by default, a deny-by-default filesystem, and a cleared environment so a sandboxed process cannot read your API keys. |

### Know what your device can actually enforce

Containment depends on kernel features that not every device has. OMERTA reports what yours can genuinely provide — **strict**, **relaxed** or **limits only** — and a request for more containment than the device can enforce is capped at what is real, and says so. It will not tell you a process is contained when it is not.

## CHAPTER 8

## Firmware Work

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The part of OMERTA with the strictest rules, because the cost of being wrong is a brick.

OMERTA can unpack and rebuild boot, vendor\_boot and dtbo images, read and decompile device trees, handle Android sparse images and take apart a super.img. What governs all of it is an evidence rule.

### Evidence labels

| Label     | Meaning                                                      |
|-----------|--------------------------------------------------------------|
| CONFIRMED | Extracted from the actual firmware or kernel in front of it. |
| LIKELY    | Strongly supported by evidence, not directly read.           |
| INFERRED  | Deduced from related facts. Treat as a hypothesis.           |
| UNKNOWN   | Not established. It stays UNKNOWN.                           |

**An UNKNOWN is never promoted to a guess to keep a task moving.** If a value cannot be extracted, OMERTA tells you what it would need in order to know, rather than supplying a plausible number.

### Standing rules

- The only copy of stock firmware is never modified. Work happens on a copy.
- A newly built boot image is not flashed immediately — it is inspected, compared and explained first.
- Flashing is a HIGH RISK action and always asks, on every policy setting.
- Every irreversible step states what it will do, to which partition, and what recovery looks like if it fails.

## CHAPTER 9

## Memory, Learning and Forgetting

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What it keeps, where it keeps it, and how to make it let go.

### Three kinds of memory

| Kind                | What it is                                  | Scope              |
|---------------------|---------------------------------------------|--------------------|
| Conversation        | The current chat, unbounded and untruncated | One chat           |
| Facts & preferences | Things you told it to remember              | Project, or shared |
| Learned documents   | Files you uploaded in the LEARN tab         | Project            |

### Forgetting is real

Removing a fact, a document or a learned behaviour deletes it and confirms the deletion. This is worth stating because it was once broken: the store returned a status string instead of the row identifier, so every removal silently did nothing while reporting success. It is now tested, including that a deletion propagates to your other devices and is not resurrected by the next sync.

### Multi-device sync

Devices exchange memory directly — phone to laptop to workstation — with no server in between. Safety travels with it: a denial recorded on one device is inherited by the others, so a refusal you set on your phone is not quietly absent on your laptop. Re-syncing is idempotent, malformed bundles are rejected rather than merged, and oversized ones are refused before they are parsed.

## CHAPTER 10

## Privacy and Security

What leaves your device, what cannot reach it, and what to be careful about.

### What leaves the device

| Situation               | What is sent, and where                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------|
| Cloud provider selected | Your prompt and the relevant context, to that provider only. Nothing to us — there is no "us" in the path. |
| Local provider selected | Nothing leaves your network.                                                                               |
| Offline mode            | Nothing leaves the device at all. Verified by a test that fails if the network is touched.                 |

### Loopback and the local-only rule

The server binds to loopback. If you expose it on your network, a token is required — and the token check cannot be fooled by a forged X-Forwarded-For, X-Real-IP or Forwarded header.

Four surfaces refuse *any* request that did not originate on this device, even with a valid token: the terminal, the editor's writes, the sandbox runner and loading a file by absolute path. They do not route through the approval gate, because they are you operating your own machine — which is precisely why they may not be driven from another one.

This was not theoretical. Each of the three was found by exploiting it: a remote caller ran `id` as root through the sandbox endpoint, overwrote a file through the editor endpoint, and read `/etc/hostname` through the learning endpoint. All three are closed, the fix is enforced in both servers immediately after authentication, and the regression is tested.

### Keys and private chats

- API keys are stored on the device and never written into logs, commits or exports.
- A sandboxed process gets a cleared environment, so it cannot read your keys even if it goes looking.
- A key can be scoped to one project. That scope is thread-local, not an environment variable: turns for different projects run at the same time, and a key placed in the process environment would be visible to all of them — which is the exact thing scoping is for.
- Private chats are encrypted with a key derived from your passcode (PBKDF2-HMAC-SHA256, then AES-GCM). The passcode is not stored. Lose it and the chat is unreadable — by anyone, including you.

### Pairing a second device

Pairing is the one route in this program that answers without a token, and only while you have an offer open on the device that already has the data. Outside that window it does not answer at all.

- Single use, and it expires — two minutes by default.
- Eight characters from an alphabet with no 0/O or 1/I in it. That is about 40 bits: weak as a password, ample for one attempt-capped window, which is all it is.
- Five wrong codes close the offer.

- A claim must come from a private address with nothing forwarding for it. Pairing is two devices in the same room; a proxy in the path means it is not.
- The offer lives in memory and is never written to disk, so a restart cancels it rather than leaving a live path to the token.

What this does *not* do is prove which device claimed. Anything on your network that guesses the code inside the window gets the token, and the mitigations above are the whole defence. It is a convenience over reading a token aloud with the blast radius kept small — not an authenticated pairing protocol, and it is not described as one.

## The audit chain

Sealing the log computes a digest for every entry that includes the digest before it, so a line cannot be altered or removed without breaking every digest after it. The head is a single string you can write down somewhere this device cannot reach.

This is tamper **evidence**, not tamper proofing, and the app says so. The key lives on the same device as the log, so anything able to rewrite the log can also recompute the chain. What it cannot do is change the head you already wrote down elsewhere — comparing that later is the entire value here. Calling it "signed" would promise something local storage cannot deliver.

## The panic wipe

Chats, memory, the learning shelf, attachments, keys, the audit log. It is deliberately **not** behind the approval gate: a panic action that stops to ask is a panic action that does not work. The confirmation is a phrase typed in full, at the moment.

Nothing is backed up first — doing that automatically would defeat the point. And it reports what it could *not* remove rather than claiming success: a file held open or a permission it lacks is listed, because pretending those were wiped is the single worst lie this program could tell.

## CHAPTER 11

# Troubleshooting

The things most likely to go wrong, and what they mean.

| Symptom                                              | Cause and fix                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chat does nothing, or says no provider is configured | No model is connected. See Chapter 2 — you need an API key or a local model server.                                                                                                                                                                                                                                                                             |
| "offline mode needs a local model"                   | You are in offline mode with no local server reachable. Start Ollama or llama.cpp and set its address, or switch to online.                                                                                                                                                                                                                                     |
| Terminal warns about job control                     | Fixed in 1.3.0. If you still see it, you are running an older build — check the stamp at the top of Settings.                                                                                                                                                                                                                                                   |
| A command is "not found" in the terminal             | Check the banner shows a tool count. If it says the toolset is unavailable it gives the reason. Note there is no python3, adb or sudo — see Chapter 6 and Chapter 12.                                                                                                                                                                                           |
| wget warns about TLS certificates                    | Expected. BusyBox's TLS does not validate them. Ask the agent to fetch instead when it matters.                                                                                                                                                                                                                                                                 |
| A .deb installs but is "unconfigured"                | Use <code>apt install ./file.deb</code> , not <code>dpkg -i</code> , so dependencies are pulled.                                                                                                                                                                                                                                                                |
| An update lost my chats                              | The app was uninstalled rather than updated, or the new build was signed with a different key. Install over the top; never uninstall.                                                                                                                                                                                                                           |
| "Backend didn't come up"                             | The launch screen now names the reason underneath. Tap <b>RETRY &amp; SHOW DIAGNOSTICS</b> : it re-runs the launch (which often succeeds on its own) and, if it fails again, prints the device, ABI, payload state and full Python traceback, with <b>COPY</b> and <b>SHARE</b> buttons. Send that text — it is the whole diagnosis, no cable or logcat needed. |
| First launch sits on "unpacking Python" for minutes  | Expected once, on slower or 32-bit devices: the app is writing a full Python standard library to private storage. The file counter moves the whole time. It never repeats until you update the app.                                                                                                                                                             |
| Sandbox says containment is limited                  | Your kernel lacks the namespaces. The report is accurate — it is telling you the truth rather than overstating protection.                                                                                                                                                                                                                                      |

## CHAPTER 12

## What Is Missing, and What Should Come Next

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An honest list. These are the gaps between what OMERTA is today and what it is trying to be, in the order they are worth closing.

### 1. An inference engine in the APK

**The biggest gap by far.** Today the phone needs either an API key or a model server elsewhere on your network. Bundling a llama.cpp server — packaged the same way the BusyBox binary is — and letting you load a GGUF file would make OMERTA genuinely self-contained: no key, no internet, no second machine. A 3B model fits comfortably on a modern handset. Everything needed to do this has now been proved to work; it is the single change that would most alter what OMERTA is.

### 2. A real python3 on the command line

Same mechanism, second use. A standalone CPython as a native binary would give the terminal a genuine interpreter, and with it pip, scripts, and most of what people actually open a terminal on a phone to do.

### 3. git and ssh

The agent understands git deeply, but the terminal has no git binary. Static builds of git, OpenSSH and curl-with-real-TLS would close the three most obvious remaining gaps in the toolset — and curl would fix the certificate-validation caveat in Chapter 6.

### 4. adb over TCP

USB-attached devices are out of reach without Android's USB Host API and a user-granted permission, so a plain adb binary cannot work. But `adb connect host : 5555` over the network is an ordinary TCP protocol and is implementable. libimobiledevice has the same USB problem with no TCP equivalent, so it is the one item on your list that is genuinely out of reach on an unrooted phone.

### 5. Signing the release properly

Builds are currently signed with the Android debug key, which is what keeps updates installing over each other. Before you publish anything, generate a real release keystore, back it up in more than one place, and sign with it. **If you lose that key you can never update the app again** — every existing install would have to be removed and reinstalled, losing user data. This is the single most irreversible decision in shipping an Android app.

### Smaller things worth having

- **Voice input** — dictating to an agent on a phone is far more natural than typing at it.
- **A share target** — send a file, a URL or a code snippet to OMERTA from any other app.
- **Scheduled and triggered runs** — a task that wakes on a schedule, or when a repository changes.
- **An encrypted backup** of everything (chats, projects, learned files, themes) to a single file you control.
- **Per-project approval policy** — trusted inside a scratch project, strict against production.

- **A cost meter** for cloud providers, so a long agent run cannot surprise you.
- **Model presets** — a cheap fast model for routine work and a strong one for hard problems, switched per task rather than globally.
- **Tablet and landscape layout** — the interface currently assumes a phone.

## CHAPTER 13

## Reference

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Settings, endpoints and files.

### Environment settings

| Variable               | Effect                                 |
|------------------------|----------------------------------------|
| OMERTA_MODE            | online   offline   auto                |
| OMERTA_PROVIDER        | Which provider to use, or auto         |
| OMERTA_APPROVAL_POLICY | ask   build   trusted                  |
| OMERTA_TERM_GUARD      | 0 disables the terminal deny-guard     |
| OMERTA_ISOLATE_NET     | Allow network inside the sandbox       |
| OMERTA_LOCAL_MODEL     | Model name for Ollama                  |
| OMERTA_OLLAMA_HOST     | Address of your Ollama server          |
| OMERTA_LLAMACPP_HOST   | Address of your llama.cpp server       |
| OMERTA_DATA_DIR        | Where chats, memory and settings live  |
| OMERTA_TOKEN           | API token when serving beyond loopback |

### HTTP endpoints

| Endpoint         | Purpose                                           |
|------------------|---------------------------------------------------|
| GET /api/version | Which build this is — version, commit, build time |
| GET /api/status  | Providers, policy, mode, toolset, memory, skills  |
| POST /api/chat   | One turn, through the approval gate               |
| /api/term/*      | Terminal. Local-only.                             |
| /api/ws/*        | Editor and workspace. Writes are local-only.      |
| /api/scratch/*   | Sandbox. Local-only.                              |
| /api/learn/*     | Learning shelf. Loading by path is local-only.    |

### Command line

| Command      | Does                               |
|--------------|------------------------------------|
| omerta       | Interactive agent in the terminal  |
| omerta serve | Start the server and web interface |

| Command                 | Does                                       |
|-------------------------|--------------------------------------------|
| omerta doctor           | Check the installation and report problems |
| omerta sync <host:port> | Sync memory with another device            |
| omerta --version        | Version, commit and build time             |

## Where your data lives

| Platform      | Path                                   |
|---------------|----------------------------------------|
| Android       | /data/data/com.omerta.agent/files/data |
| Linux / macOS | ~/.omerta (or \$OMERTA_DATA_DIR)       |
| Windows       | %USERPROFILE%\\.omerta                 |

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OMERTA AI v1.16.0 — Owner's Manual. Generated from the repository, so the version on the cover is the version of the build that produced it. Silence Is The Only Unbreakable Code.